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# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

## MGU-BVOC (HONOURS) Regular EXAMINATION April 2026

## SECOND SEMESTER

**Skill Development Components (SDC) - MG2SDCREM100 - ENERGY AND POWER FUNDAMENTALS**

**(2025 ADMISSION ONWARDS)**

**Duration: 2 Hours Maximum Marks: 70 Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(1) and Appreciation(Ap)**

**Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.**

## Part A

### Short answer

### Answer any 25 questions

### Each question carries 1 mark.

**1. Identify the purpose of power generation in a power system [K]/[CO1]**

**Ans:**The purpose of power generation in a power system is to produce electrical energy from other forms of energy (like thermal, hydro, or renewable sources) to supply electricity to consumers.

**2. State the purpose of power transmission. [K]/[CO1]**

**Ans:**The purpose of power transmission is to transfer electrical energy from generating stations to distribution centers over long distances efficiently and with minimal losses.

**3. Classify the type of value measured by an AC voltmeter (Peak or RMS) [K]/[CO1]**

**Ans:**An AC voltmeter measures the RMS (Root Mean Square) value of the voltage.

**4. State Ohm's Law.** **[K]/[C01]**

**Ans:**Ohm's Law states that the current flowing through a conductor is directly proportional to the voltage across it, provided the temperature and other physical conditions remain constant.

Mathematically:V=IR.

**5. State the unit of active power.** **[K]/[C01]**

**Ans:**The unit of active power is watt (W).

**6. Describe RMS value of AC voltage.** **[K]/[C01]**

**Ans:**RMS value of AC voltage is the value of a DC voltage that produces the same heating effect as the AC voltage over one complete cycle.

**7. Identify the components required to calculate utilization factor.** **[U] / [CO2]**

**Ans:** The components required to calculate utilization factor are:

- Maximum demand
- Connected load

Utilization factor = Maximum demand / Connected load

**8. Identify examples of inductive loads** **[U]/[CO2]**

**Ans:** Examples of inductive loads are:

- Electric motors
- Transformers

**9. State the unit of load factor.** **[U]/[CO2]**

**Ans:** No unit (dimensionless).

**10. Identify the components required to calculate demand factor. [U] / [CO2]**

**Ans:** To calculate demand factor, the required components are:

- Maximum demand
- Total connected load

**11. State the range of power factor. [U] / [CO2]**

**Ans:** The power factor ranges from 0 to 1 (or 0 to unity).

**12. Interpret why reactive power is associated with capacitive loads. [U] / [CO2]**

**Ans:**Reactive power is associated with capacitive loads because they store energy in an electric field and return it to the source, causing current to lead voltage without consuming real power.

**13. Differentiate between the first block and second block in block rate tariff.**

**Ans:** First block: Initial units of electricity are charged at a higher rate.Second block: Subsequent units are charged at a lower rate to encourage higher consumption.

**[An] / [CO3]**

**14. Interpret why reactive power is absent in a resistive load. [U] / [CO2]**

**Ans**: Reactive power is absent in a resistive load because voltage and current are in phase, so all the power supplied is real power with no energy stored or returned.

**15. Examine the billing pattern in two-part tariff. [An] / [CO3]**

**Ans:** Fixed charge (based on connected load or maximum demand)and Energy charge (based on units of electricity consumed)

**16.Compare smart billing and conventional billing systems. [An] / [CO3]**

**Ans:** Smart billing: Uses digital/smart meters with automatic, real-time recording and billing.

Conventional billing: Uses manual meter reading with periodic (monthly) billing.

**17. Differentiate between analog and digital energy meters. [An] / [CO3]**

**Ans:**Analog energy meter: Uses mechanical parts (like a rotating disc) to measure energy.

Digital energy meter: Uses electronic circuits and displays readings digitally with higher accuracy.

**18. Differentiate between cost-based tariff and incentive-based tariff. [An] / [CO3]**

Ans:Cost-based tariff: Charges are set based on the actual cost of supplying electricity.

Incentive-based tariff: Charges are designed to encourage or discourage certain usage patterns through discounts or penalties.

**19. Compare analog energy meter and digital energy meter. [An] / [CO3]**

**Ans:**Analog energy meter: Mechanical, uses a rotating disc and pointer.

Digital energy meter: Electronic, displays readings digitally with higher accuracy and no moving parts.

**20. Why are government policies necessary for improving energy efficiency? [E] / [CO4]**

**Ans:** Government policies are necessary because they set standards, enforce regulations, and encourage efficient energy use, leading to reduced energy consumption and environmental protection.

21.**Why is star rating important while purchasing appliances? [E] / [CO4]**

**Ans:**. Star rating indicates the **energy efficiency of an appliance**, helping consumers choose devices that **consume less electricity and reduce energy bills**.

**22. Justify the need for energy efficiency standards in India. [E] / [CO4]**

**Ans:** Energy efficiency standards are needed in India to reduce energy consumption, lower electricity costs, conserve resources, and reduce environmental pollution.

**23.Why should industries adopt renewable energy systems? [E] / [CO4]**

**Ans**: Industries should adopt renewable energy systems to reduce dependence on fossil fuels, lower operating costs, and minimize environmental pollution.

**24. Why is benchmarking important in energy audit? [E] / [CO4]**

**Ans:** Benchmarking is important in an energy audit because it allows you to compare a facility's energy performance with standards or similar facilities, helping to identify inefficiencies and potential areas for energy savings.

**25.Why is load management necessary for reducing peak demand? [E] / [CO4]**

**Ans:** Load management is necessary because it shifts or reduces electricity usage during peak hours, thereby lowering peak demand and reducing energy costs and strain on the power system.

**26. Why is walk-through audit suitable for small-scale industries? [E] / [CO4]**

**Ans:**A walk-through audit is suitable for small-scale industries because it is quick, simple, and low-cost, allowing basic identification of energy-saving opportunities without detailed analysis.

**27.Why is conservation better than increasing power generation? [E] / [CO4]**

**Ans:** Conservation is better than increasing power generation because it reduces energy demand at lower cost and with less environmental impact than producing additional power.

**[1X25=25]**

## Part B

Short Essay

Answer any **5** questions

Each question carries **5** marks

**28. State the importance of distribution system in supplying power to consumers.** **[K] / [CO1**

**Ans**: The distribution system is crucial because it delivers electrical power from transmission networks to end users safely and efficiently. Its importance includes:

- It ensures reliable and continuous supply of electricity to homes, industries, and commercial users.
- It steps down high transmission voltage to usable levels for consumers.
- It helps maintain power quality, such as proper voltage and minimal losses.
- It enables wide coverage, supplying power to urban as well as rural areas.
- It provides control, protection, and metering, ensuring safe operation and proper billing.

**29. Explain different forms of power in electrical systems.** **[K] / [CO1]**

**Ans:** In electrical systems, power is classified into different forms based on how energy is used and transferred in a circuit.

- ❖ Active power (real power) is the actual power that performs useful work, such as running motors, lighting lamps, and heating devices. It is measured in watts (W) and represents the energy consumed by resistive components.
- ❖ Reactive power is the power that oscillates between the source and reactive components like inductors and capacitors. It does not perform useful work but is necessary for maintaining magnetic and electric fields in the system. It is measured in volt-ampere reactive (VAR).
- ❖ Apparent power is the total power supplied to the circuit and is the combination of active and reactive power. It is measured in volt-amperes (VA) and represents the overall demand on the electrical system.

These three forms are related through the power triangle, where apparent power is the vector sum of active and reactive power. Understanding these forms helps in improving efficiency and power factor in electrical systems.

**30. Explain the importance of improving power factor in electrical systems. [U] / [CO2]**

**Ans:**Improving power factor is important in electrical systems because it increases the efficiency of power utilization. A higher power factor means more of the supplied power is converted into useful work, reducing wastage of energy.

It helps in reducing current flow in the system for the same amount of useful power. Lower current results in reduced $I^2R$ losses in cables and equipment, thereby improving overall system efficiency.

An improved power factor also reduces voltage drops in the system, leading to better voltage regulation and improved performance of electrical equipment.

Additionally, it decreases the load on generators, transformers, and transmission lines, allowing them to carry more useful load without overloading.

Finally, improving power factor helps in reducing electricity costs, as many utilities impose penalties for low power factor, making it economically beneficial for consumers.

**31. Compare monthly load curve and annual load curve [U] / [CO2]**

**Ans:**A monthly load curve represents the variation of electrical load over a period of one month. It shows daily or weekly fluctuations in demand within that month, helping to analyze short-term consumption patterns, peak demand periods, and operational performance. It is useful for planning day-to-day or seasonal load management and identifying immediate energy-saving opportunities.

In contrast, an annual load curve represents the variation of load over an entire year. It provides a long-term view of how demand changes across different months and seasons, reflecting factors like weather changes, industrial activity, and consumer behavior.

While the monthly load curve focuses on short-term analysis and operational control, the annual load curve is used for long-term planning, such as capacity expansion, maintenance scheduling, and forecasting future energy requirements.

Additionally, the annual load curve helps in understanding overall system utilization and load factor, whereas the monthly load curve gives more detailed insight into specific periods of high or low demand within a month.

Thus, both curves are important, with the monthly load curve aiding short-term decisions and the annual load curve supporting long-term planning and system development.

**32. Analyze the working principle of Time of Day tariff with a suitable example.**

**[An] / [CO3]**

**Ans:** The Time of Day (ToD) tariff is a pricing mechanism in which the cost of electricity varies depending on the time at which it is consumed. The main principle is to charge higher tariffs during peak demand hours and lower tariffs during off-peak periods. This encourages consumers to shift their energy usage to times when the demand on the power system is lower.

The working principle is based on load management. During peak hours, when many users consume electricity simultaneously, the tariff is increased to discourage excessive use and reduce system stress. During off-peak hours, when demand is low, the tariff is reduced to promote better utilization of available generation capacity.

For example, a utility may set the tariff as ₹8 per kWh from 6 PM to 10 PM (peak hours), ₹6 per kWh during normal hours, and ₹4 per kWh from 10 PM to 6 AM (off-peak hours). An industrial consumer can reduce electricity costs by operating heavy machinery during off-peak hours instead of peak hours.

Thus, the ToD tariff helps in reducing peak demand, improving load factor, optimizing generation, and lowering overall energy costs for both utilities and consumers.

**33. Compare maximum demand tariff and two-part tariff. [An] / [CO3]**

**Ans**: The maximum demand tariff is based on the highest level of power demand recorded during a billing period, along with the energy consumed. Consumers are charged not only for the total energy used (kWh) but also for their maximum demand (kW or kVA), which reflects the peak load they place on the system. This type of tariff encourages consumers to control their peak demand to avoid higher charges.

The two-part tariff consists of two separate components: a fixed charge and a variable charge. The fixed charge is based on connected load or maximum demand (kW/kVA), covering the cost of infrastructure and service, while the variable charge depends on the actual energy consumed (kWh).

In comparison, maximum demand tariff specifically emphasizes the measurement and billing of the highest demand recorded, often requiring demand meters, whereas two-part tariff is a broader structure that always includes both fixed and energy charges.

Maximum demand tariff mainly focuses on reducing peak load on the system, while two-part tariff ensures recovery of both fixed and running costs of electricity supply.

Thus, both tariffs involve demand consideration, but maximum demand tariff highlights peak

usage penalties, whereas two-part tariff balances fixed and variable cost recovery.

**34. Evaluate the importance of improving power factor in industrial systems.**

**[E] / [CO4]**

**Ans:** Improving power factor is highly important in industrial systems because most industrial loads (like motors and transformers) are inductive and tend to operate at low power factor. A low power factor means more current is required to deliver the same amount of useful power, leading to inefficiency.

By improving the power factor, the current drawn from the supply decreases for the same load. This reduces $I^2R$ losses in cables, transformers, and other equipment, thereby increasing overall system efficiency and reducing energy wastage.

An improved power factor also leads to better voltage regulation. Lower current results in reduced voltage drops across the system, ensuring that machinery operates at proper voltage levels, which enhances performance and lifespan of equipment.

It also reduces the burden on electrical infrastructure such as generators, transformers, and transmission lines. This allows industries to use the existing system capacity more effectively without needing costly upgrades.

Finally, improving power factor has economic benefits. Industries can avoid penalties imposed by utilities for low power factor and may even receive incentives, resulting in lower electricity bills and improved cost efficiency

**[5x5 = 25]**

## Part C

Essay

Answer any **2** questions

Each question carries **10** marks

**35. Explain the charging and discharging process of a capacitor in detail. [K] / [CO1]**

**Ans:**A capacitor is an electrical device that stores energy in the form of an electric field. The charging and discharging processes explain how it stores and releases this energy in a circuit.

Charging of a capacitor:
When a capacitor is connected to a DC supply through a resistor, it begins to charge. Initially, at the moment of connection, the capacitor behaves like a short circuit, allowing maximum current to flow. Electrons start accumulating on one plate, while an equal positive charge builds up on the other plate. As the charge increases, a voltage develops across the capacitor plates, opposing the supply voltage.

Discharging of a capacitor:
When the charged capacitor is disconnected from the supply and connected across a resistor, it starts to discharge. The stored energy is released, and current flows in the opposite direction.

During discharge, the current is maximum at the beginning and gradually decreases to zero as the capacitor loses its stored charge. After about 5 time constants, the capacitor is considered fully discharged.

**36.Explain in detail the daily load curve and its significance in power system operation. [U] / [CO2]**

**Ans:**A daily load curve is a graphical representation of the variation of electrical load (power demand) on a power system over a 24-hour period. The load is plotted on the Y-axis (in kW or MW), and time is plotted on the X-axis (hours of the day). It shows how the demand for electricity changes throughout the day depending on consumer activities.

In a typical daily load curve, the load is low during late night and early morning hours, as most residential and commercial activities are minimal. As the day begins, the load gradually increases due to domestic usage, commercial operations, and industrial activities. The curve usually reaches a peak during evening hours when residential demand (lighting, cooking, appliances) combines with other loads. After the peak period, the load again decreases as activities reduce at night.

The curve clearly indicates base load, average load, and peak load. Base load represents the

minimum level of demand that is always present, while peak load is the maximum demand during the day. The area under the curve represents the total energy consumed during the day.

The significance of the daily load curve in power system operation is very important. It helps in planning generation schedules, ensuring that enough power is produced to meet demand at all times. It also assists in selecting suitable types of power plants, such as base load plants (like thermal or nuclear) and peak load plants (like hydro or gas turbines).

Additionally, the daily load curve is useful for load management and demand-side control, helping utilities reduce peak demand through methods like load shifting and Time of Day tariffs. It also aids in improving the load factor, which indicates the efficiency of system utilization.

Furthermore, it helps in maintenance planning, economic operation, and forecasting future demand patterns. By analyzing the curve, engineers can make decisions to ensure reliability, reduce costs, and improve overall efficiency of the power system.

Thus, the daily load curve is an essential tool for understanding demand patterns and ensuring efficient and reliable operation of electrical power systems.

**37.Examine how a good tariff encourages efficient energy consumption. [An] / [CO3]**

**Ans:**A tariff is the pricing structure used by electricity suppliers to charge consumers for the electrical energy they consume. A good tariff system is designed not only to recover the cost of generation, transmission, and distribution, but also to encourage efficient and responsible use of electricity. It plays an important role in influencing consumer behavior and improving overall power system efficiency.

A good tariff encourages efficient energy consumption by making electricity pricing reflect the actual cost of supply. For example, higher charges during peak demand periods and lower charges during off-peak periods motivate consumers to shift their usage, thereby reducing unnecessary load on the system.

It also promotes load management and peak demand reduction. Tariff structures such as Time of Day (ToD) tariffs or peak pricing discourage excessive consumption during peak hours. This helps in flattening the load curve and improving the load factor of the system.

Another important aspect is that a good tariff encourages energy conservation awareness among consumers. When consumers see higher bills for higher usage, they tend to adopt energy-efficient practices such as using efficient appliances, reducing wastage, and optimizing usage patterns.

A well-designed tariff also supports fair cost recovery and economic efficiency. By properly distributing fixed and variable charges (as in two-part tariffs), it ensures that consumers who use more power or impose higher demand on the system pay accordingly, promoting fairness and discouraging overuse.

In industrial systems, tariffs based on maximum demand or power factor incentives/penalties encourage industries to improve power factor and reduce peak load. This leads to reduced losses and better utilization of electrical infrastructure.

Good tariff systems also help utilities in system planning and reliability improvement. When demand is more evenly distributed, utilities can operate generation units more efficiently, reduce the need for additional peak power plants, and minimize operational stress.

Additionally, tariffs with slab-wise pricing (increasing block rates) discourage excessive consumption by charging higher rates for higher usage levels. This ensures basic energy needs are met at lower cost while discouraging wastage.

Overall, a good tariff acts as a demand-side management tool, guiding consumers toward efficient electricity use while ensuring financial sustainability of the power system.

Thus, a well-structured tariff system is essential for promoting efficient energy consumption, reducing peak demand, improving system performance, and ensuring economic and environmental benefits.

**38.Assess the importance of each stage in the energy audit procedure. [E] / [CO4]**

**Ans:** An energy audit is a systematic process of analyzing energy use in a facility to identify opportunities for energy conservation and efficiency improvement. It is carried out in stages, and each stage plays a crucial role in ensuring accurate results and effective implementation of energy-saving measures.

The first stage is preliminary data collection and planning. In this stage, basic information such as utility bills, plant layout, production data, and energy consumption patterns is collected. This stage is important because it helps the auditor understand the scope of the audit and identify major energy-consuming areas before detailed analysis begins.

The second stage is a walk-through or preliminary survey. Here, a quick inspection of the facility is carried out to identify obvious energy wastage, inefficient equipment, and operational issues. This stage is important because it provides a quick overview of energy performance and helps in deciding whether a detailed audit is required.

The third stage is detailed energy audit and data measurement. In this stage, instruments are used to measure parameters such as voltage, current, power factor, temperature, and fuel consumption. This stage is very important because accurate measurements form the basis for identifying real energy losses and inefficiencies.

The fourth stage is analysis of collected data. The measured data is analyzed to calculate energy consumption patterns, load factors, efficiency levels, and losses in the system. This stage is important because it converts raw data into meaningful information for

decision-making and helps identify areas with high energy-saving potential.

The fifth stage is identification of energy conservation opportunities (ECOs). In this stage, specific areas for improvement such as motor replacement, power factor correction, lighting upgrades, or process optimization are identified. This stage is important because it directly leads to practical solutions for reducing energy consumption.

The sixth stage is economic evaluation of proposed measures. Here, the cost-benefit analysis, payback period, and return on investment are calculated. This stage is important because it ensures that recommended measures are financially viable and acceptable to management.

The final stage is reporting and implementation monitoring. A detailed audit report is prepared with recommendations, and implementation is monitored for effectiveness. This stage is important because it ensures that identified energy-saving measures are actually applied and verified for performance improvement.

Thus, each stage of the energy audit procedure is essential, from data collection to implementation, as they collectively ensure accurate assessment, practical solutions, and effective energy management in a facility.

**[10x2 = 20]**

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